

Diagnosing and Mitigating Sycophancy and Skepticism in LLM Causal Judgment

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Abstract

Do frontier LLMs reason causally, or do they pattern-match, yielding under pressure and hedging under uncertainty? We frame causal judgment as evaluation along three axes, *Utility*, *Safety*, and *Wise Refusal*, across Pearl’s Ladder. We introduce *Recursive Causal Audit* (RCA), a process-integrity evaluator whose Judge has *no access to gold labels*: it checks whether a model’s answer is entailed by its own derivation, internally consistent, and not dominated by user hints under pressure. RCA unifies persona and pressure: prompt tone is the intervention that regulates pressure-induced drift. For fine diagnostic resolution we use CAUSALT3,¹ with explicit trap families and standardized pressure protocols. CAUSALT3 reveals a *Skepticism Trap* (Claude Haiku rejects 60% of valid L1 links) and a *Scaling Paradox* (GPT-5.2 underperforms GPT-4-Turbo by 55 points on L3, driven by paralysis rather than hallucination). Under RCA, operating points shift toward the high-Utility, high-Safety quadrant without retraining, consistent with much of the observed failure arising from how answers are rendered under pressure rather than from missing causal knowledge.

rates, achieving high specificity by sacrificing sensitivity. Second, a *Scaling Paradox*: on ambiguous counterfactuals, larger models can *regress* by defaulting to paralysis rather than engaging the reasoning they demonstrably possess. These are not edge cases but recurring behavioral patterns that interact with alignment and prompting pressure.

To study these failures, we frame causal judgment as an evaluation problem along three axes: *Utility* (sensitivity), *Safety* (specificity), and *Wise Refusal* (calibrated abstention) across Pearl’s three levels of causation: L1 Association, L2 Intervention, and L3 Counterfactuals (Pearl, 2009; Pearl and Mackenzie, 2018). We then introduce *Recursive Causal Audit* (RCA), an evaluator of process integrity that tests whether a model’s final label is actually supported by its own derivation. *Crucially*, RCA’s Judge has *no access to gold labels*: it verifies trace-output consistency, internal coherence, and resistance to hint dominance under pressure, rather than checking correctness against an answer key. RCA further unifies *persona and pressure*: the same tonal interventions that induce sycophancy or self-doubt in evaluation are inverted inside RCA as a control mechanism against pressure-induced drift. To support this evaluator-centered analysis at fine diagnostic resolution, we develop CAUSALT3,¹ a curated benchmark spanning Pearl’s three levels with expert-annotated trap families and standardized pressure protocols.

Under RCA, model operating points shift toward the high-Utility, high-Safety quadrant because the final answer must remain faithful to the model’s own structured reasoning trace. This matters especially under pressure: RCA is designed not merely to retry, but to regulate how answers are rendered when social pressure, epistemic pressure, or ambiguity would otherwise distort the output. The results are consistent with the hypothesis that much of the observed pathology reflects output-layer biases rather than missing causal knowledge.

1 Introduction

Can frontier LLMs reason causally, or do they merely pattern-match, agreeing with users when pressured and hedging when uncertain? Current evaluations obscure the answer: aggregate accuracy conflates genuine capability with refusal, hedging, and pressure-induced drift, making it difficult to distinguish failures of reasoning from failures of output behavior.

This paper diagnoses two pathologies in LLM causal judgment. First, a *Skepticism Trap*: safety-tuned models reject valid causal links at alarming

¹CAUSALT3 is a 454-instance diagnostic subset of CAUSALT5K (Geng et al., 2026).

Table 1: **Terminology and label mapping.** Top: metric names. Bottom: level-specific label sets.

Metric	Standard Equiv.	Informal
Utility	Sensitivity / TPR	Sheep
Safety	Specificity / TNR	Wolf
Wise Refusal	Calib. abstention	n/a
Level	Label set	Abstention
L1 (Assoc.)	YES / NO	AMBIGUOUS
L2 (Interv.)	VALID / FLAWED	AMBIGUOUS
L3 (Counterf.)	VALID / INVALID	CONDITIONAL

1.1 Utility vs. Safety in Causal Judgment

We decompose causal accuracy into two orthogonal dimensions, with a third calibration axis for underdetermined cases:

- *Utility* (Sensitivity): the true positive rate on valid claims (*Sheep*).
- *Safety* (Specificity): the true negative rate on invalid claims (*Wolves*).
- *Wise Refusal*: correct abstention on ill-posed cases. The corresponding label is level-specific: AMBIGUOUS at L1/L2, CONDITIONAL at L3.

Table 1 maps these to standard terminology. This decomposition highlights asymmetric failure modes invisible in aggregate accuracy. A model can score well overall by being overly agreeable (high Utility, low Safety) or overly skeptical (high Safety, low Utility).

1.2 Key Findings

Our evaluation yields five findings:

1. *The Skepticism Trap (L1)*. While L1 Safety is near-ceiling for frontier models, Utility can collapse for safety-tuned models. Claude 3.5 Haiku rejects 60% of valid associational claims (40% Utility), consistent with systematic over-refusal.
2. *The Sycophancy Trap (L2)*. Under social pressure, correct rejections flip to endorsements. Even frontier models reversed a high percentage of FLAWED judgments when challenged, revealing pressure-induced sycophancy distinct from baseline bias.
3. *Asymmetric baseline biases*. Different models default to opposite failure profiles. Skepticism-heavy models (e.g., Claude Haiku) achieve high Safety but sacrifice Utility by rejecting valid claims. Endorsement-heavy models achieve high Utility but sacrifice Safety by accepting invalid ones. A single aggregate accuracy score masks this tradeoff entirely; only the two-axis Utility/Safety decomposition makes it visible.

4. *The L3 Scaling Paradox*. On CAUSALT3-L3, capability increases do not guarantee improved counterfactual judgment. GPT-5.2 underperforms the older GPT-4-Turbo (20% vs. 75% Safety) by defaulting to paralysis (CONDITIONAL) when faced with underspecification, an “Ambiguity Trap” that larger models appear more prone to on this benchmark.
5. *Process verification substantially mitigates diagnosed failures*. Under RCA, measured L3 operating points shift toward the high-Utility, high-Safety quadrant and CONDITIONAL rates drop, indicating that the Scaling Paradox can be substantially mitigated at inference time without retraining (Section 3).

1.3 Contributions

We make four primary contributions:

1. *RCA as a process-integrity evaluator*: an inference-time protocol whose Judge operates *without access to gold labels*, verifying schema compliance, internal consistency, trace-output consistency, and hint non-dominance under pressure. RCA further unifies persona and pressure: prompt tone is a controlled intervention that counteracts pressure-induced answer drift (Section 3).
2. *A three-axis evaluation framework for causal judgment*: Utility (sensitivity), Safety (specificity), and Wise Refusal (calibrated abstention), exposing opposing failure modes that aggregate accuracy obscures.
3. *Diagnostic findings across Pearl’s hierarchy*: the Skepticism Trap (L1 over-refusal), the Sycophancy Trap (L2 pressure-induced reversal), and the Scaling Paradox (L3 non-monotonic regression). RCA substantially mitigated them at inference time without retraining.
4. CAUSALT3: a 454-case diagnostic benchmark spanning Pearl’s hierarchy with expert-annotated trap families and standardized pressure protocols, providing the instrument needed for RCA-based evaluation.

2 Related Work

We position this work first in the literature on *evaluation of causal judgment and reasoning reliability* in LLMs, and second in the literature on benchmarks and pressure-sensitive behavioral failure. Our central question is not only whether models answer causal questions correctly, but whether their

Table 2: **Benchmark Comparison.** “Traps” = explicit causal pitfalls (confounding, collider bias, Simpson’s paradox). “Ambig.” = cases requiring calibrated uncertainty. [†]Truthfulness baseline, not a causal benchmark.

Benchmark	Levels	Traps	2-Axis	Ambig.
CLadder	L1-L3	n/a	No	No
CRASS	L3	n/a	No	No
CORR2CAUSE	L2	n/a	No	No
e-CARE	L1-L2	n/a	No	No
TruthfulQA [†]	n/a	n/a	No	No
CAUSALT3	L1-L3	Yes	Yes	Yes

final answers remain faithful to their own reasoning under ambiguity and pressure. From this perspective, RCA contributes an evaluator of process integrity, while CAUSALT3 provides the diagnostic setting needed to measure that integrity across Pearl’s hierarchy. Recent surveys highlight the potential of LLMs for causality (Kıcıman et al., 2023; Zhang et al., 2023), but debate remains: do these models possess genuine structural understanding, or do they merely act as “causal parrots” (Zečević et al., 2023)? Table 2 summarizes how this evaluator-centered view differs from prior benchmark-only formulations.

2.1 Evaluation of Causal Reasoning

Foundations and Formalism. The evaluation of causal systems has long been grounded in structural principles rather than surface plausibility alone (Spirtes et al., 2000; Schölkopf et al., 2021). Recent LLM work has extended this agenda through graph-based and benchmark-based tests of causal reasoning, but most prior evaluations still center on end-task correctness rather than process integrity under pressure. Our work complements these benchmarks by asking a different question: when a model gives a causal answer, is that answer actually supported by its own derivation, or has the final label drifted under refusal bias, ambiguity pressure, or user influence? CLadder (Jin et al., 2023) generates queries from causal graphs, effectively testing the “do-calculus” (Pearl, 2009), but real-world causal judgment often requires navigating informal ambiguity rather than formal symbols. CAUSALT3 takes a different approach, prioritizing *semantic depth* over synthetic scale. Each of the 454 expert-curated vignettes targets a specific structural failure mode (e.g., distinguishing Confounding from Mediation) embedded in natural language, and is paired with standardized pressure protocols that make pressure-sensitive drift measurable.

Pearl’s ladder benchmarks. Evaluating models against the Causal Hierarchy Theorem (CHT) (Bareinboim et al., 2022) is a growing standard. CLadder (Jin et al., 2023) generates $\sim 10K$ queries from causal graphs across L1 to L3. While its graph-first design provides formal control, synthetic variable construction yields puzzle-like scenarios distant from deployment contexts. CRASS (Frohberg and Binder, 2022) targets L3 counterfactuals but does not require explicit causal model construction. Recent work has also probed counterfactual consistency (Dehghanighobadi et al., 2025) and logical modification (Huang et al., 2024), though often without the specific focus on failure-mode decomposition (Utility vs. Safety).

Causal discovery and association. The work of CORR2CAUSE (Jin et al., 2024) tests causal direction inference from correlations, but low performance suggests directionality from correlations alone is ill-posed for LLMs. At L1 (Association), the “Reversal Curse” (Berglund et al., 2023) demonstrates that models often fail to generalize $A \rightarrow B$ to $B \rightarrow A$, a fundamental associational deficit. Our work aligns with the “Epidemiology of LLMs” perspective (Plecko et al., 2025). That work argues models may memorize variable names but lack knowledge of the underlying observational distributions ($P(X, Y)$) required to identify traps like confounding (Rubin, 1974).

Commonsense causality. e-CARE (Du et al., 2022) and BIG-Bench Causal Judgment (Srivastava et al., 2023) test narrative causal attribution. However, these often admit learned scripts rather than structural reasoning. CAUSALT3 moves beyond narrative plausibility to test specific structural identifiability conditions (Pearl, 2009).

2.2 Sycophancy and Truthfulness

The Sycophancy Problem. Sycophancy, where models agree with user biases to optimize for perceived helpfulness, is a known side-effect of Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022; Bai et al., 2022). Sharma et al. (2023) and Turpin et al. (2023) showed that models often rationalize incorrect answers when prompted with a biased context, even under Chain-of-Thought reasoning (Wei et al., 2022). This phenomenon extends to multi-turn dialogues (Hong et al., 2025) and even objective domains like theorem proving (Petrov et al., 2025). Our work extends this to the causal domain, show-